

Solutions to JEE Advanced – 6 | JEE 2022 | Paper-1

PHYSICS

1.(ABD)

- (A) For normal incidence on face AC,
- $r_1 = 30^\circ$

$$\frac{\sin i}{\sin r_1} = \sqrt{2} \Rightarrow i = 45^\circ$$

Angle made by ray with horizontal = $45^\circ - 30^\circ = 15^\circ$

- (B) For total internal reflection at AC,
- $r_2 > \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) \Rightarrow r_2 > 45^\circ$

$$\text{But } r_2 = 90^\circ - (180^\circ - (30^\circ + r_1 + 90^\circ)) = 30^\circ + r_1 \Rightarrow r_1 > 15^\circ$$

$$\Rightarrow i > \sin^{-1}(\sqrt{2} \sin 15^\circ) \Rightarrow i > \sin^{-1}((1.41)(0.26))$$

$$\Rightarrow \text{For total internal reflection at AC, } i > \sin^{-1}(0.37)$$

For vertically travelling incident ray, $i = 60^\circ$

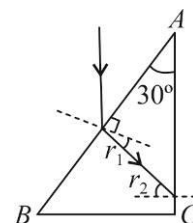
So, clearly, this ray will undergo total internal reflection at face AC

- (C) For total internal reflection at AC,
- $r_2 > 45^\circ$

$$r_1 + r_2 = 30^\circ \Rightarrow r_2 < 30^\circ$$

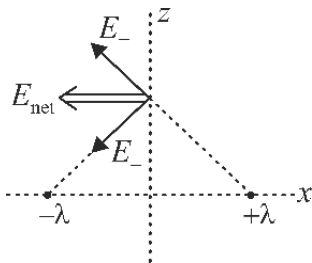
So, this ray will NOT undergo total internal reflection at face AC

- (D) For normal incidence,
- $r_1 = 0^\circ \Rightarrow r_2 = 30^\circ \Rightarrow e = 45^\circ$



2.(ABD)

- (A) As shown in the figure, at a point on the
- $+z$
- axis, field is in
- $-x$
- direction. Similarly, it can be understood that at any point on the
- $-z$
- axis, field is also in
- $-x$
- direction.



- (B) By extension, it can be understood that at any point on the
- yz
- plane, field is in
- $-x$
- direction. Since everywhere on the
- yz
- plane field is perpendicular to the plane, the
- yz
- plane is an equipotential surface

- (C) At
- $(x, 0, 0)$
- ,
- $\vec{E} = \left[\frac{\lambda}{2\pi\epsilon_0(a-x)} \right](-\hat{i}) + \left[\frac{\lambda}{2\pi\epsilon_0(a+x)} \right](-\hat{i}) = \left[\frac{\lambda a}{\pi\epsilon_0(a^2 - x^2)} \right](-\hat{i})$

- (D) Work done, $W = -(q) \int_{x=0}^{x=\frac{a}{2}} \vec{E} \cdot d\vec{x} = \int_0^{\frac{a}{2}} \frac{\lambda q a}{\pi\epsilon_0(a^2 - x^2)} dx = \frac{\lambda q}{2\pi\epsilon_0} \left[\int_0^{\frac{a}{2}} \frac{dx}{a-x} + \int_0^{\frac{a}{2}} \frac{dx}{a+x} \right]$
- $$= \frac{\lambda q}{2\pi\epsilon_0} \log_e \left| \frac{a+x}{a-x} \right|_0^{\frac{a}{2}} = \frac{\lambda q}{2\pi\epsilon_0} \log_e 3$$

3.(ABD) Current leads the voltage by a phase difference :

$$\phi = \tan^{-1} \left(\frac{\frac{1}{\omega C} - \omega L}{R} \right) = \tan^{-1} \left(\frac{\frac{1}{(100)(20 \times 10^{-6})} - (100)(0.5)}{R} \right) = 45^\circ \Rightarrow R = 450 \Omega$$

$$(A) \quad P_{av} = I_{rms}^2 R \Rightarrow I_{rms} = \sqrt{\frac{P_{av}}{R}} = \frac{1}{\sqrt{5}} A$$

$$(B) \quad \text{Power factor } \cos \phi = \cos 45^\circ = \frac{1}{\sqrt{2}}$$

$$(C) \quad V_m = \sqrt{2} I_{rms} Z = (\sqrt{2}) \left(\frac{1}{\sqrt{5}} \right) (450\sqrt{2}) = 180\sqrt{5} V$$

$$(D) \quad \text{Current lags behind the voltage source by } \frac{\pi}{4} \Rightarrow \frac{\omega L - \frac{1}{\omega C}}{R} = \tan 45^\circ = 1 \Rightarrow \omega = 1000 \text{ rad/s}$$

4.(BCD)

(A) Since the magnetic force on a particle is always in a direction perpendicular to its velocity, the work done by the magnetic force is zero. Therefore, the kinetic energy and the speed of the particle remain constant.

(B) Radius of the circular path of the particle, $r = \frac{mv}{qB}$

$$\sin \theta = \frac{d}{r} = \frac{qBd}{mv}$$

(C) Time period of the circular path, $T = \frac{2\pi m}{qB}$

$$\text{Time spent by the particle inside the region, } t = T \left(\frac{\theta}{2\pi} \right) = \frac{m\theta}{qB}$$

(D) Magnetic force on a moving charged particle, $\vec{F} = q(\vec{v} \times \vec{B})$

Since the velocity of the particle is always perpendicular to the magnetic field, the force on the particle is constant in magnitude. The direction of the force keeps changing as the particle follows a circular path, since the force is always perpendicular to the velocity.

5.(BCD) We can see using the right-hand rule that:

(a) at points inside Loop 1, magnetic field is directed into the plane, and

(b) at points outside Loop 1, magnetic field is directed out of the plane

Therefore, flux through Loop 2 is always out of the plane.

If Loop 2 is moved away, the field becomes weaker, and thus the flux decreases. So, an anti-clockwise current is induced.

Conversely, if Loop 2 is moved closer, a clockwise current is induced.

If Loop 2 is moved along a circular path around the centre of Loop 1, the flux through Loop 2 does not change because of symmetry. Hence, there is no current induced in Loop 2.

6.(ABD)

$$\text{Radius of an orbit, } r \propto n^2 \Rightarrow r_1 > r_2$$

$$\text{Angular momentum of the electron in an orbit, } L = n \frac{h}{2\pi} \Rightarrow L_1 > L_2$$

$$\text{Total energy of the electron in an orbit, } E \propto -\frac{1}{n^2} \Rightarrow E_1 > E_2$$

As energy difference between successive orbits decreases as n increases, so $\lambda_1 > \lambda_2$

7.(ABD) Let currents in ammeters A_1 & A_2 be I_1 & I_2 respectively (both towards right). Then,

$$E_1 - E_2 - I_1 R_1 = 0 \Rightarrow I_1 = \frac{E_1 - E_2}{R_1}$$

$$E_2 - I_2 R_2 = 0 \Rightarrow I_2 = \frac{E_2}{R_2}$$

If $E_1 > E_2$, I_1 & I_2 can be same

$$\text{If } E_2 > E_1 \Rightarrow I_1 = \frac{E_2 - E_1}{R_1} \text{ towards left}$$

$\Rightarrow I_1$ & I_2 will be same for a value of $R_1 < R_2$

8.(ABCD)

(A) At steady state, potential difference across $C_1 = \left(\frac{3}{2+3}\right)(30) = 18V$

(B) Charge on $C_2 = (30 - 18)(3) = 36\mu C$

(C) Heat = Work done by battery – Potential energy stored in capacitors

$$= (36)(30) - \frac{1}{2} \left(\frac{6}{5}\right)(30)^2 = 540\mu J$$

(D) Now, when the switch is shifted to position 2, current flows only in the right loop

$$\Rightarrow i = \frac{12}{4} = 3A$$

9.(C) Fringe width originally, $\beta_1 = \frac{4.8 \times 10}{6} = 8mm$

Fringe width finally, $\beta_2 = \frac{4.8 \times 10}{8} = 6mm$

Fringe width, $\beta = \frac{D\lambda}{d} \Rightarrow \Delta\beta = \frac{\lambda}{d} \Delta D \Rightarrow \lambda = d \left(\frac{\Delta\beta}{\Delta D} \right) = (10^{-4}) \left(\frac{2 \times 10^{-3}}{0.5} \right) m = 400nm$

10.(D) Distance between the central maxima and the closest point of complete darkness, $d = \frac{\beta_2}{2} = 3mm$

11.(C) $\frac{dR}{dT} = R_0 \alpha \Rightarrow \int_{R_0}^R dR = R_0 \alpha \int_{T_0}^T dT \Rightarrow R = R_0 (1 + \alpha(T - T_0))$

At the equilibrium temperature, the rate of heat generation in the rod due to the flow of current through it is equal to the rate of heat loss to the surroundings. So,

$$\frac{V_0^2}{R_0} = k(T_0 - T_S) \quad \text{and} \quad \frac{V^2}{R_0(1 + \alpha(T - T_0))} = k(T - T_S)$$

Dividing the equations,

$$\frac{V_0^2(1 + \alpha(T - T_0))}{V^2} = \frac{T_0 - T_S}{T - T_S}$$

$$\Rightarrow \left(\frac{100}{120}\right)^2 (1 + \alpha(540 - 480)) = \left(\frac{480 - 300}{540 - 300}\right) \Rightarrow \alpha = 1.33 \times 10^{-3} K^{-1}$$

$$12.(B) \quad \frac{\left(\frac{V_0^2}{R_0}\right)}{\left(\frac{V^2}{R}\right)} = \frac{T_0 - T_S}{T - T_S} \Rightarrow \frac{V_0 i_0}{Vi} = \frac{T_0 - T_S}{T - T_S} \Rightarrow \frac{i_0}{i} = \left(\frac{480 - 300}{540 - 300}\right) \left(\frac{120}{100}\right) = 0.9 \Rightarrow i = \frac{4.5}{0.9} = 5A$$

1.(18) Let the angle of refraction in the m^{th} layer be r_m

$$\frac{\sin i}{\sin r_1} = \frac{\mu_1}{1.5}, \quad \frac{\sin r_1}{\sin r_2} = \frac{\mu_2}{\mu_1}, \quad \frac{\sin r_2}{\sin r_3} = \frac{\mu_3}{\mu_2} \quad \dots \quad \frac{\sin r_{m-1}}{\sin r_m} = \frac{\mu_m}{\mu_{m-1}}$$

Multiplying all the equations,

$$\frac{\sin i}{\sin r_m} = \frac{\mu_m}{1.5} \quad \text{and similarly,} \quad \frac{\sin i}{\sin r_{m+1}} = \frac{\mu_{m+1}}{1.5}$$

The ray is able to penetrate into the m^{th} layer but not into the $(m+1)^{th}$ layer

$$\Rightarrow r_m < 90^\circ \quad \text{and} \quad r_{m+1} > 90^\circ$$

$$\Rightarrow \sin r_m < 1 \quad \text{and} \quad \sin r_{m+1} > 1$$

[Note: Do not get worried about the inequality $\sin r_{m+1} \geq 1$, it is just a mathematical way of writing that the angle of incidence is greater than or equal to the critical angle.]

$$\text{So,} \quad \left(\frac{1.5}{\mu_m}\right) \sin i < 1 \quad \text{and} \quad \left(\frac{1.5}{\mu_{m+1}}\right) \sin i > 1$$

$$\Rightarrow \mu_m > \frac{6}{5} \quad \text{and} \quad \mu_{m+1} < \frac{6}{5}$$

$$\Rightarrow \frac{3}{2} - \frac{m}{61} > \frac{6}{5} \quad \text{and} \quad \frac{3}{2} - \frac{m+1}{61} < \frac{6}{5}$$

$$\Rightarrow m < 18.3 \quad \text{and} \quad m > 17.3 \quad \Rightarrow \quad m = 18$$

2.(1) At $t = t_0$, the number of nuclei of type A, $A_2 = A_1 e^{-\frac{t_0}{T} \log_e 2}$

$$\text{At } t = t_0, \text{ the number of nuclei of type B, } B_2 = B_1 + A_1 \left(1 - e^{-\frac{t_0}{T} \log_e 2}\right)$$

$$\text{Let } e^{-\frac{t_0}{T} \log_e 2} = p$$

$$\text{Dividing the two equations,} \quad \frac{A_2}{B_2} = \frac{A_1 p}{B_1 + A_1(1-p)}$$

$$\Rightarrow \frac{A_2}{B_2} = \frac{\frac{A_1}{B_1} p}{1 + \frac{A_1}{B_1}(1-p)} \Rightarrow \frac{1}{5} = \frac{\frac{p}{2}}{1 + \frac{1}{2}(1-p)} \Rightarrow p = \frac{1}{2} \Rightarrow \frac{t_0}{T} = 1$$

$$3.(6) \quad B = \frac{\mu_0 qV \sin \theta}{4\pi r^2} \\ = 10^{-7} \cdot \frac{0.5 \times 5 \times 10^3 \times \sin(37^\circ)}{5^2} = 6\mu T$$

$$4.(0.14) \quad i(t) = \frac{V}{R} \left(1 - e^{-\frac{R}{L}t} \right) \quad \text{and} \quad i_{\max} = \frac{V}{R}$$

$$i(t) = \frac{3}{4} i_{\max} \Rightarrow t = \frac{2L}{R} \log_e 2 = 1.4 \times 10^{-4} \text{ s} = 0.14 \text{ ms}$$

5.(11.25)

Potential energy of a dipole kept in a uniform field, $U = -\vec{p} \cdot \vec{E} = -pE \cos \theta$, where θ is the angle between the dipole moment vector and the electric field vector.

Work done in rotating a dipole kept in a uniform electric field is

$$U_f - U_i = (-pE(-1)) - (-pE(1)) = 2pE = 2(p) \left(\frac{\sigma}{2\epsilon_0} \right) = 11.25 \text{ J}$$

$$6.(5) \quad \text{Number of photons striking the plate per second} = \frac{I(A \sin 30^\circ)}{\left(\frac{hc}{\lambda} \right)} = \frac{(500)(10^{-3})}{e \left(\frac{1242}{414} \right)} = \frac{1}{6e}$$

$$\text{Photoelectric current, } i = \left(\frac{1}{6e} \right) \left(\frac{3}{100} \right) e = 5 \text{ mA}$$

CHEMISTRY

1.(ABCD)

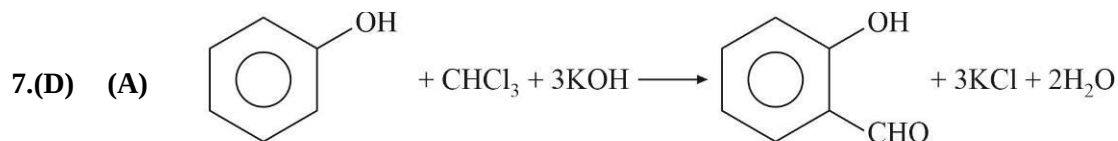
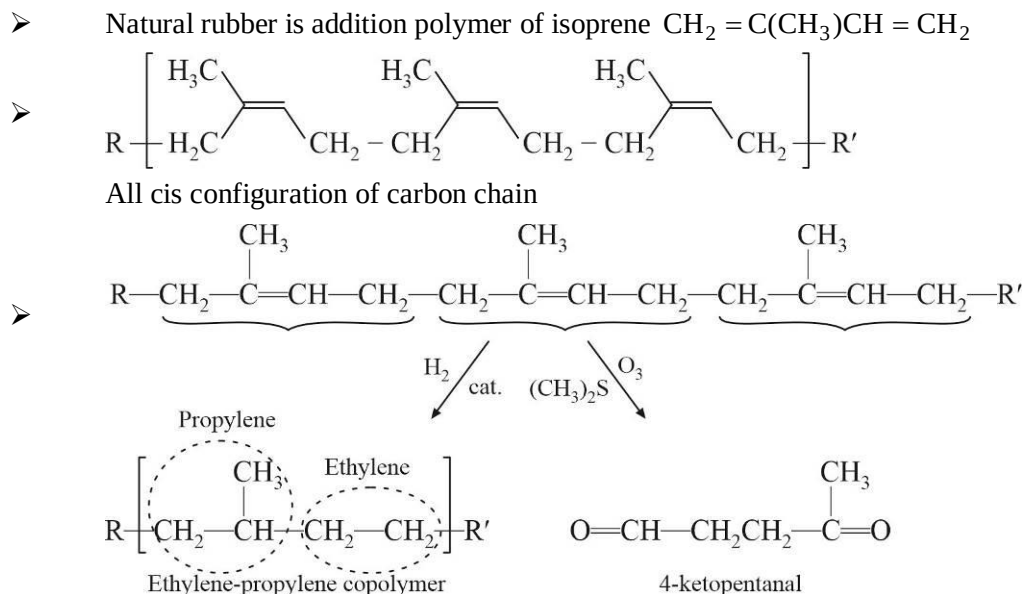
- (A) Zinc uranyl acetate, $\text{ZnUO}_2(\text{CH}_3\text{COO})_4$, is used as a laboratory reagent in the determination of sodium concentration of solution. In this, precipitate of sodium zinc uranyl acetate (yellow) $(\text{NaZn}(\text{UO}_2)_3(\text{CH}_3\text{COO})_9)$ is formed
- (B) Alkaline K_2HgI_4 (Nessler's reagent) is used for confirmatory test of the ammonium ion
 $2\text{K}_2\text{HgI}_4 + 4\text{KOH} + \text{NH}_4\text{Cl} \longrightarrow \text{HgOHgNH}_2\text{I} + 7\text{KI} + \text{KCl} + 3\text{H}_2\text{O}$
- (C) $3\text{K}_4[\text{Fe}(\text{CN})_6] + 4\text{FeCl}_3 \longrightarrow \text{Fe}_4[\text{Fe}(\text{CN})_6]_3 + 12\text{KCl}$
Prussian blue
- (D) $2\text{FeSO}_4 \cdot 7\text{H}_2\text{O} \xrightarrow{\Delta} \text{Fe}_2\text{O}_3 + \text{SO}_2 + \text{SO}_3 + 7\text{H}_2\text{O}$

2.(ACD)

- | | | | |
|-----|--|---|----------------|
| (A) | | Close loop of delocalizable $6\pi^-$ | (Aromatic) |
| (B) | | Absence of close loop of delocalizable $6\pi^-$ | (Not aromatic) |
| (C) | | Close loop of delocalizable $6\pi^-$ | (Aromatic) |
| (D) | | Close loop of delocalizable $6\pi^-$ | (Aromatic) |

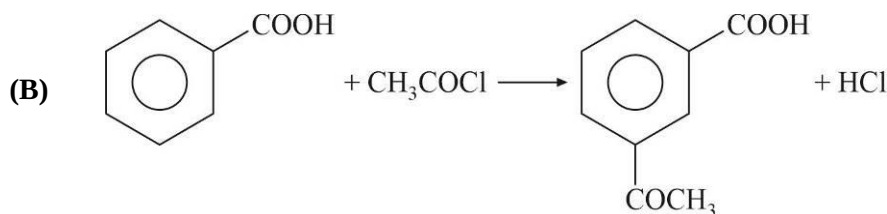
- 3.(A) (A) $\Rightarrow$ Ionization isomers : Produce different ions on ionization
 (B) $\Rightarrow$ Hydrate isomers : Different number of H_2O as ligand and solvent of crystallization
 (C) $\Rightarrow$ Linkage isomers : Ambident ligand
 (D) $\Rightarrow$ Coordination isomers : Different type of coordination sphere
- 4.(AB) Germanium is group-14 element and on doping with P produce electron rich semiconductor that is n-type and there will be an increase in conductivity due to presence of free electron.
- 5.(ABD) The initially adsorbed layer can't act as a substrate for further adsorption because Langmuir adsorption is monolayer. All other statements are correct.

6.(ABCD)



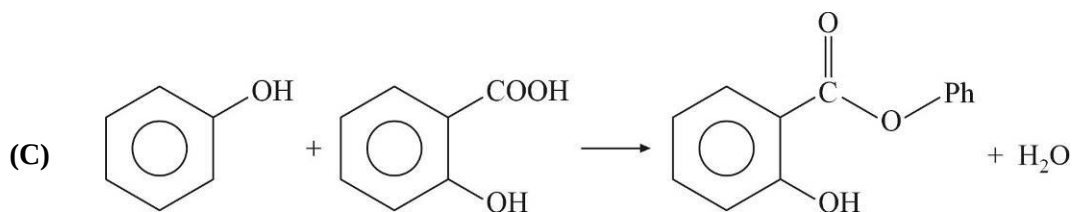
Reimer-Tiemann reaction

Not aspirin



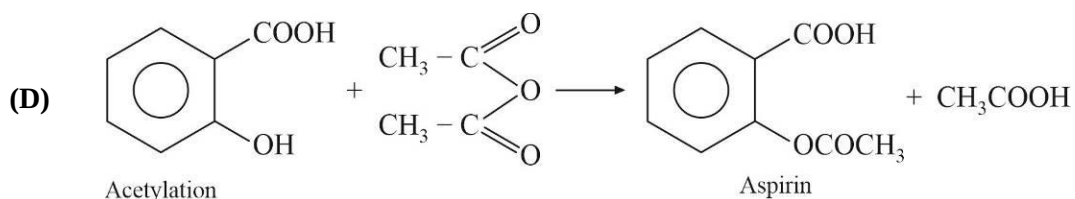
F.C. acylation

Not aspirin



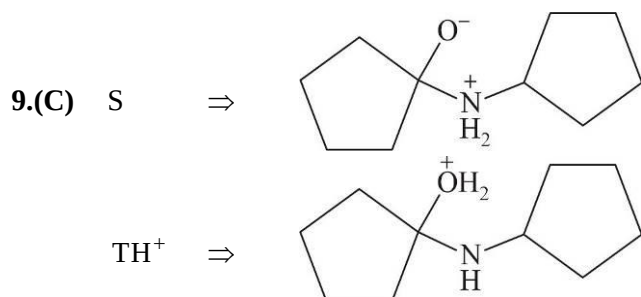
Esterification

Not aspirin

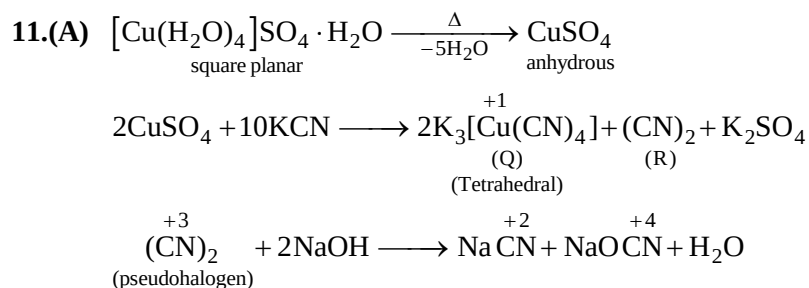


8.(ABCD)

- BaCl_2 salt impart green colour in fireworks
- Compounds of boron imparts green colour in flame test
- Malachite is green coloured basic copper carbonate, $\text{CuCO}_3 \cdot \text{Cu(OH)}_2$
- Emerald is mineral beryl ($\text{Be}_3\text{Al}_2(\text{SiO}_3)_6$) coloured green by trace amounts of chromium Cr^{3+} ion in octahedral void of ccp lattice of oxide ion. Green colour is due to d-d transition.



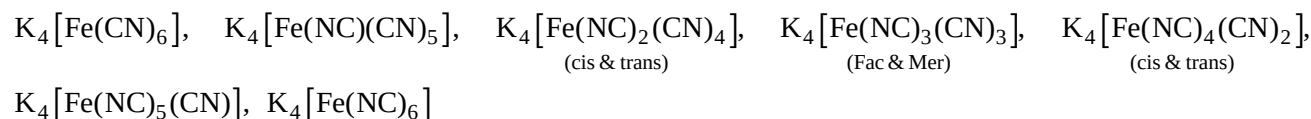
10.(D) With increase in H^+ concentration equilibrium K_2 shift towards forward direction according to Le-chatelier's principle and amine will also be protonated significantly.



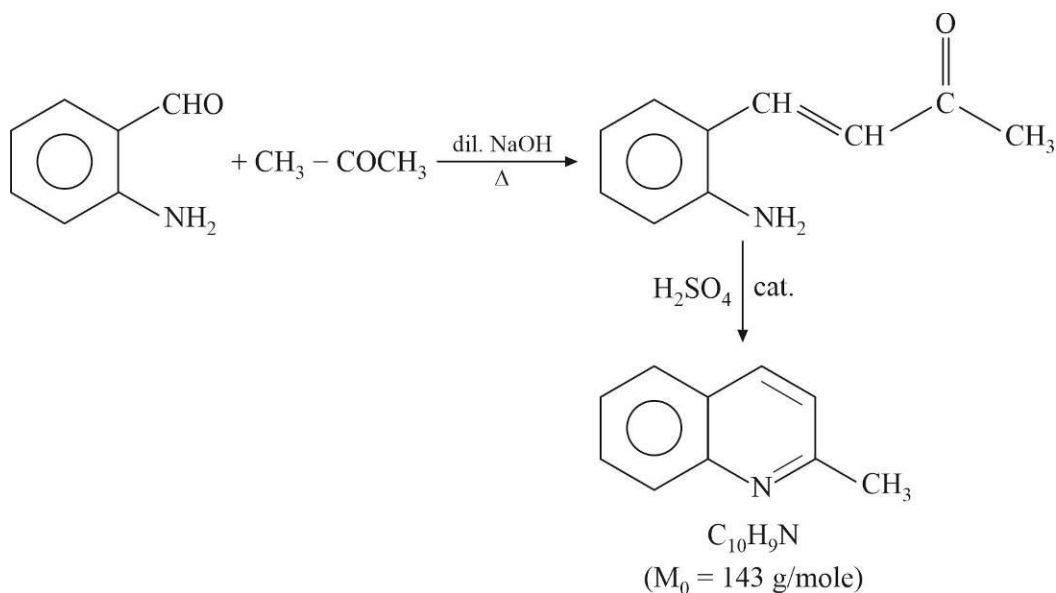
In above reaction carbon undergoes oxidation as well as reduction hence called disproportionation reaction.

12.(A) Blue coloured aq. copper sulphate solution absorb orange-red coloured light.

1.(10) Total ten isomers are possible for octahedral complex having formula $\text{Fe}(\text{CN})_2 \cdot 4\text{KCN}$.

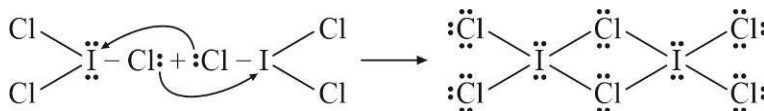


2. (9.79) First reaction between o-amino benzaldehyde and acetone is cross aldol condensation in presence of NaOH. Resulting product undergoes intramolecular nucleophilic addition and elimination of amine group across $\text{C}=\text{O}$ to form cyclic imine $\text{C}_{10}\text{H}_9\text{N}$ having molecular mass 143.

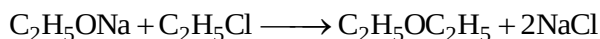
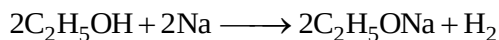


$$\% \text{ of N} = \frac{14}{143} \times 100 = 9.79\%$$

3.(20) I_2Cl_6 is dimer of ICl_3 .



4.(7.47) Ethylchloride and ethanol both can react with sodium but hydrogen (H_2) is obtained only from reaction of ethanol with sodium.



Mole of $\text{H}_2 = \frac{1}{2} \times \text{mole of } \text{C}_2\text{H}_5\text{OH}$; $P = 1\text{ atm}$, $V = 200\text{ mL}$, $T = 300\text{ K}$ [$1\text{ atm} = 760\text{ mmHg}$]

$$\frac{1 \times 200}{1000 \times 0.0821 \times 300} = \frac{1}{2} \times \frac{W}{46} \Rightarrow W = \frac{1 \times 200 \times 2 \times 46}{1000 \times 0.0821 \times 300} = 0.7470$$

$$\% \text{ of ethanol} = \frac{0.7470}{10} \times 100 = 7.47\%$$

5.(1) In first row of d-block elements only Cu^{2+}/Cu electrode has positive value of standard reduction potential.

6.(5) All are sulphides except Cuprite (Cu_2O), Siderite (FeCO_3) and Kaolinite ($\text{Al}_2(\text{OH})_4\text{Si}_2\text{O}_5$).

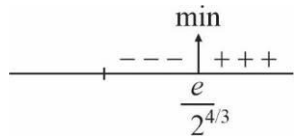
$$\begin{aligned}
 1.(ACD) \quad & \left(\frac{1}{x\sqrt{1-x^2}} - \frac{\sin^{-1} x}{x^2} \right) dx = 2xydx + x^2 dy \\
 \Rightarrow & \left(\frac{x \frac{1}{\sqrt{1-x^2}} - \sin^{-1} x}{x^2} \right) dx = \int d(x^2 y) \Rightarrow \int d\left(\frac{\sin^{-1} x}{x}\right) = \int d(x^2 y) \\
 \Rightarrow & \frac{\sin^{-1} x}{x} = x^2 y + C \Rightarrow y = \frac{\sin^{-1} x}{x^3} - \frac{C}{x^2} \\
 \Rightarrow & \lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x^3} - \frac{C}{x^2} = \frac{1}{6} \Rightarrow C = 1 \Rightarrow f(x) = \frac{\sin^{-1} x}{x^3} - \frac{1}{x^2}
 \end{aligned}$$

Clearly, even function

$$f(-1) = f(1) = \frac{\pi}{2} - 1$$

$$\begin{aligned}
 2.(BCD) \quad & f'(x) = 2(2x \ln(2x) - 2x) - (x \ln x - x) = 4x \ln(2x) - x \ln x - 3x = x(\ln(16x^4) - \ln x - 3) \\
 \Rightarrow & f'(x) = x(\ln(16x^3) - 3)
 \end{aligned}$$

$$\text{Now } f'(x) > 0 \Rightarrow \ln(16x^3) > 3 \Rightarrow 16x^3 > e^3 \Rightarrow x > \frac{e}{2^{4/3}}$$



Clearly $f(x)$ has local minima at $x = \frac{e}{2^{4/3}}$ but no local maxima.

$$3.(ABC) \quad \sin^2 x + 2 \sin x + \frac{11}{4} \geq 2$$

$$\sin x \geq -\frac{1}{2}$$

$$4.(ABD) \quad a < p \Rightarrow \frac{a}{p} < 1, \text{ limit} = 0$$

$$a > p, \frac{a}{p} > 1, \text{ limit} \rightarrow \infty$$

$$a = p, l = 0$$

$$\lim_{x \rightarrow \infty} \left(\frac{ax^2 + bx + c}{px^2 + qx + r} \right)^{mx+n} = e^{(b-q)m/p}$$

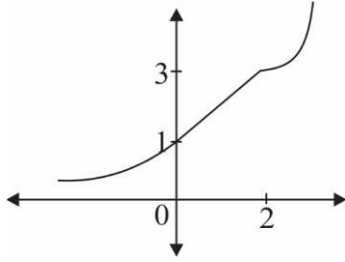
$$5.(AB) \quad \int \frac{\cos^{-1} \frac{x}{a}}{x^2} dx = -\frac{\pi}{2} \frac{1}{x} - \int \frac{\sin^{-1} \frac{x}{a}}{x^2} dx$$

$$x = a \cos \theta$$

$$\Rightarrow -\frac{1}{a} \int \frac{\theta}{\cos^2 \theta} \sin \theta d\theta = -\frac{1}{a} \int \theta \tan \theta \sec \theta d\theta$$

$$= -\frac{1}{a} (\theta \sec \theta - \ln(\sec \theta + \tan \theta)) + c = -\frac{1}{x} \cos^{-1} \left(\frac{x}{a} \right) + \frac{1}{a} \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right) + c$$

6.(ABC) $f(0^-) = f(0^+) \Rightarrow \lambda = 1$



7.(BC) For same number probability is $\frac{{}^8C_4}{{}^9C_4} \times \frac{1}{{}^8C_4}$

$$P(\text{A's 4 digit number} > \text{B's}) = \frac{{}^8C_3 \times {}^8C_4 + \frac{1}{2} \left(({}^8C_4)^2 - {}^8C_4 \right)}{{}^9C_4 \times {}^8C_4}$$

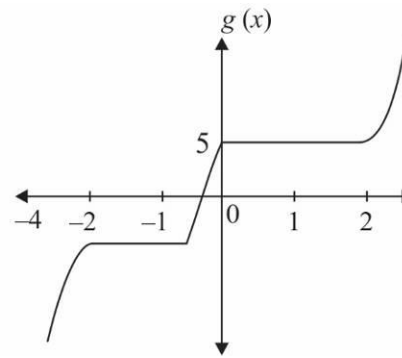
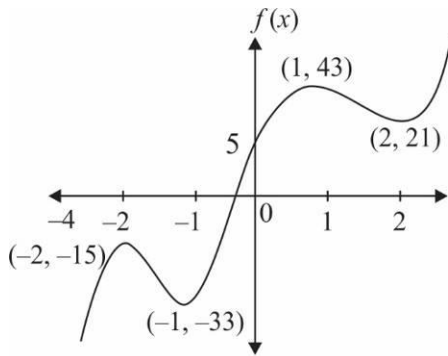
8.(ABCD) $A + B = ABAB = A^2 + A$

$$\Rightarrow B = A^2$$

$$BAB = A^5 = A + I \Rightarrow A(A^4 - I) = I$$

$$B^5 - A^5 = (A^5)^2 - A^5 = (A + I)^2 - (A + I) = A^2 + A = A + B$$

9.(B)



10.(D) $-33 < -\lambda < -15$ or $21 < -\lambda < 43 \Rightarrow \lambda \in (-43, -21) \cup (15, 33)$

11.(D) Equation of L is $\frac{x-1}{4} = \frac{y-1}{-3} = \frac{z-1}{-1}$

12.(D) Required line passes through $(1, 1, 1)$ and parallel to $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$

1.(1) $f(x+2y) = f(x) + f(2y) + e^{x+2y} (x+2y) - xe^x - 2ye^{2y} + 4xy \quad \forall x, y \in \mathbb{R}$

Put $x = 0, y = 0$ so, $f(0) = 0$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\{f(x) + f(h) + e^{x+h}(x+h) - xe^x - he^h + 2hx\} - \{f(x) + f(0) + e^x x - xe^x - 0 + 0\}}{h}$$

$$= f'(0) + xe^x + e^x - 1 + 2x$$

$$\therefore f'(0) = 1 \Rightarrow \text{so, } f'(x) = 2x + (x+1)e^x \Rightarrow f(x) = x^2 + xe^x + c$$

$$\therefore f(0) = 0 \Rightarrow \text{so, } f(x) = x^2 + xe^x \Rightarrow f(1) - e = 1$$

$$2.(5) \quad \int_0^2 \frac{375x^2}{(1+x^2)^4} dx = 2^n ; \int_0^2 \frac{375x^5}{x^8(1+x^{-2})^4} dx = 2^n ; \int_0^2 \frac{375x^{-3}}{(1+x^{-2})^4} dx = 2^n$$

$$\text{Simplifying } \frac{375}{6} \left(\frac{x^2}{1+x^2} \right)^3 \Bigg|_0^2 = 2^n$$

$$\frac{375}{6} \cdot \frac{4.4.4}{5.5.5} = 2^n \Rightarrow 2^n = 32 \Rightarrow n = 5$$

$$3.(4) \quad |A| + \left| \frac{1}{2}A \right| + \left| \frac{1}{4}A \right| + \left| \frac{1}{8}A \right| + \dots \infty = |A| + \frac{1}{2}|A| + \frac{1}{2^2}|A| + \frac{1}{2^3}|A| + \dots = \frac{|A|}{1 - \frac{1}{2}} = 2|A|$$

4.(3) The man has to win at least 4 times
 $\therefore$ the required probability

$$\begin{aligned} &= {}^7C_4 \left(\frac{1}{2} \right)^4 \cdot \left(\frac{1}{2} \right)^3 + {}^7C_5 \cdot \left(\frac{1}{2} \right)^5 \left(\frac{1}{2} \right)^2 + {}^7C_6 \left(\frac{1}{2} \right)^6 \cdot \frac{1}{2} + {}^7C_7 \left(\frac{1}{2} \right)^7 \\ &= ({}^7C_4 + {}^7C_5 + {}^7C_6 + {}^7C_7) \cdot \frac{1}{2^7} = \frac{64}{2^7} = \frac{1}{2} \end{aligned}$$

$$5.(4) \quad f(x) = \begin{cases} 0 & x=1 \\ -\tan \pi x & 0 < x < 1 \\ \tan \pi x & x > 1 \end{cases}$$

$$\lim_{x \rightarrow 1} f(x) = 0$$

6.(3) Using LH Rule